Deep SQL EDA Report

Spider Text-to-SQL Dataset: Complete EDA & Profiling Report

Generated from local artifacts only — 166 schemas, 166 SQLite DBs, 1034 gold SQL examples (dev_gold.sql).

1. Introduction

This report documents exploratory data analysis (EDA) and database profiling of the local Spider Text-to-SQL artifacts. Given a natural language question and a relational schema, the task is to generate executable SQL. Analysis focuses on SQL structural complexity, schema linking difficulty (JOINs / foreign keys), table/column distributions, and data quality.

All tables, charts, and percentages below are computed by generate_milestone2_report.py from schema.json, database/*.sqlite, and local *_gold.sql files — not handwritten statistics.

2. Objectives

- Validate local Spider schemas and SQLite databases
- Profile every database/table/column (rows, NULL rates, cardinality)
- Quantify SQL complexity from available gold SQL
- Analyze JOIN types, FK alignment, and relation topology
- Export reproducible CSV finding tables + charts + this document

3. Dataset Identification

Dataset: Spider — Large-Scale Human-Labeled Dataset for Complex and Cross-Domain Text-to-SQL (Yu et al., EMNLP 2018).

Local scope: train+dev style SQLite folder + schema.json + gold SQL present in this directory.

Task scope: Text-to-SQL only.

3.1 Local Inputs Used

Artifact	Role	Count
schema.json	Schema metadata (tables, columns, types, PK/FK)	166
database/	Executable SQLite databases	166
*_gold.sql	Gold SQL queries for complexity / JOIN EDA	1034
eda_csvs/	Exported finding tables	see §16

	(generated)	

4. Dataset Structure

File / Folder	Role
schema.json	Normalized schema metadata for train+dev DBs
database/<db_id>/<db_id>.sqlite	Executable DBs for profiling & execution eval
dev_gold.sql (and any other *_gold.sql)	query \t db_id gold labels
eda_csvs/*.csv	Generated table & JOIN analysis exports
figures/*.png	Generated charts

5. Gold SQL Record Format

Each gold SQL line is tab-separated: <SQLQuery>\t<db_id>

6. Split Analysis (from local gold SQL)

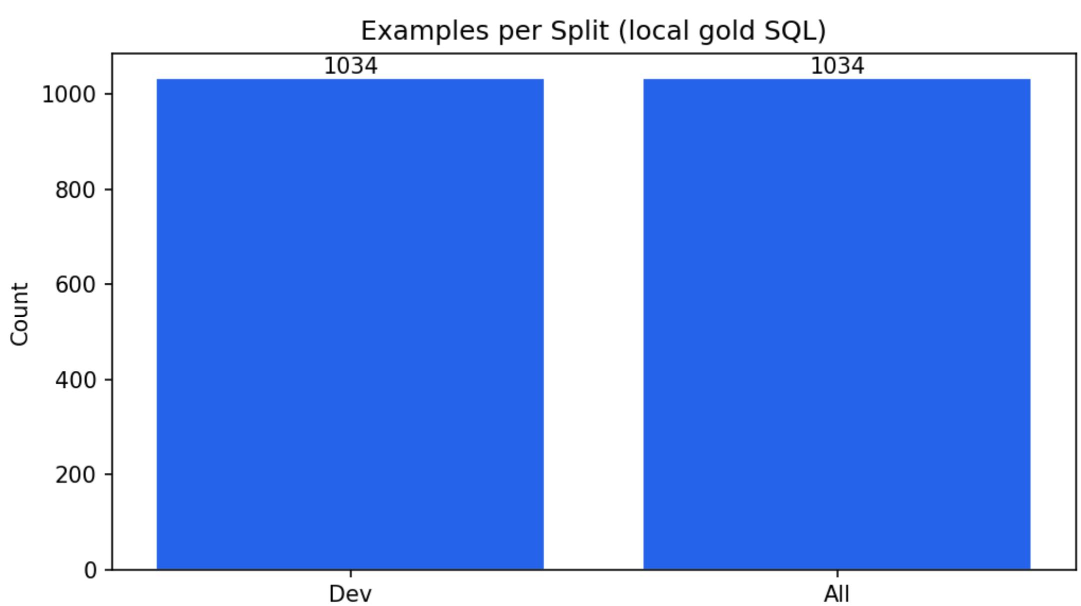

Figure 1: Examples per available gold split

Split	Examples	Unique DBs	Avg SQL tokens	Median SQL tokens
dev	1034	20	15.62	13.0

Combined gold examples analysed: 1034. Unique databases referenced: 20.

7. SQL Query Complexity Analysis

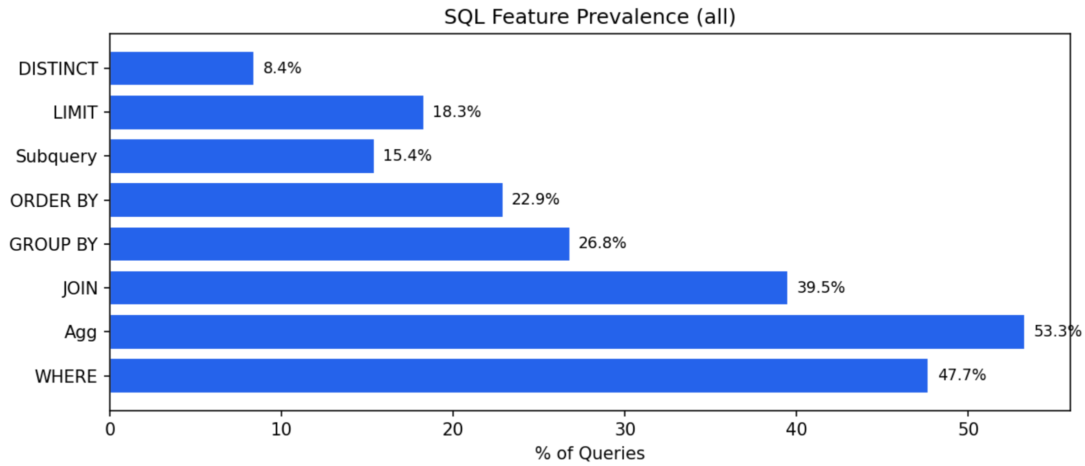

Figure 2: SQL feature prevalence

SQL Feature	Dev %
WHERE clause	47.7
Aggregation	53.3
JOIN	39.5
GROUP BY	26.8
ORDER BY	22.9
Subquery	15.4
LIMIT	18.3
DISTINCT	8.4
HAVING	7.6
LIKE	1.4
BETWEEN	0.8
IN	4.8
INTERSECT	3.9
EXCEPT	3.0
UNION	1.1

7.1 Aggregation Function Breakdown

Function	Count
COUNT	428
AVG	71
MAX	42
SUM	35
MIN	27

7.2 Query Complexity Tiers

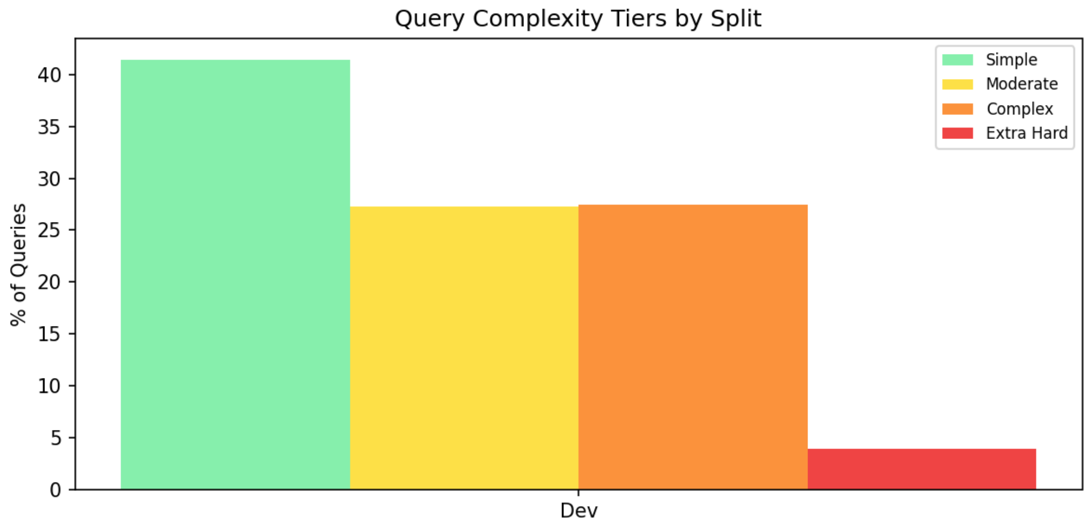

Figure 3: Complexity tiers

Tier	Dev
Simple	428 (41.4%)
Moderate	282 (27.3%)
Complex	284 (27.5%)
Extra Hard	40 (3.9%)

7.3 JOIN Distribution

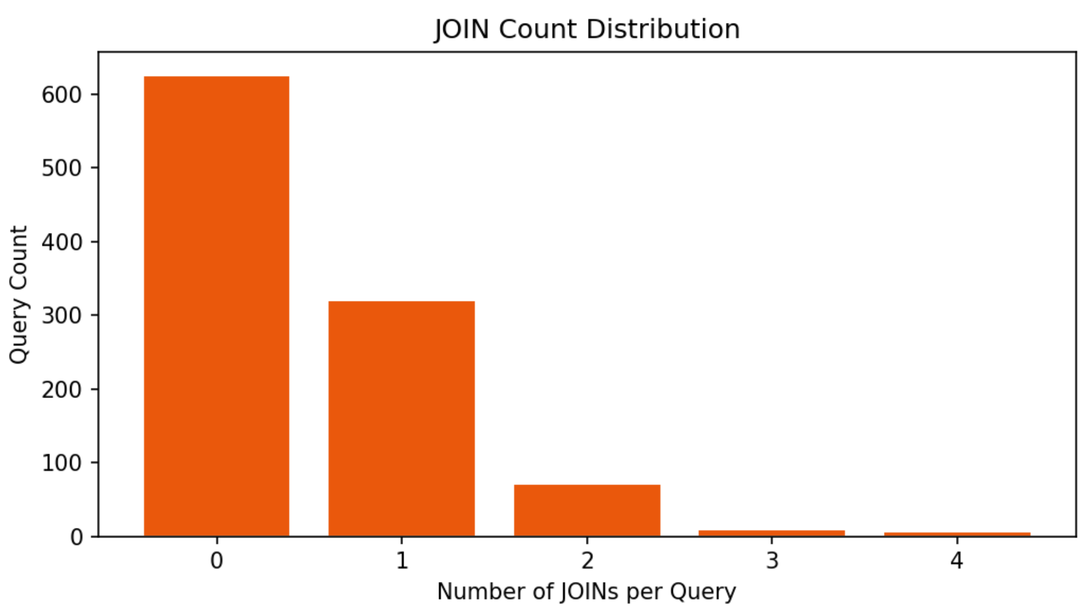

Figure 4: JOINs per query

Distribution: 0 JOINs: 626 (61%), 1 JOINs: 320 (31%), 2 JOINs: 72 (7%), 3 JOINs: 10 (1%), 4 JOINs: 6 (1%).

8. Natural Language Question Analysis

Local gold SQL files do not include natural-language questions. Question-pattern EDA requires train/dev JSON; this run substitutes SQL hardness as a structural proxy.

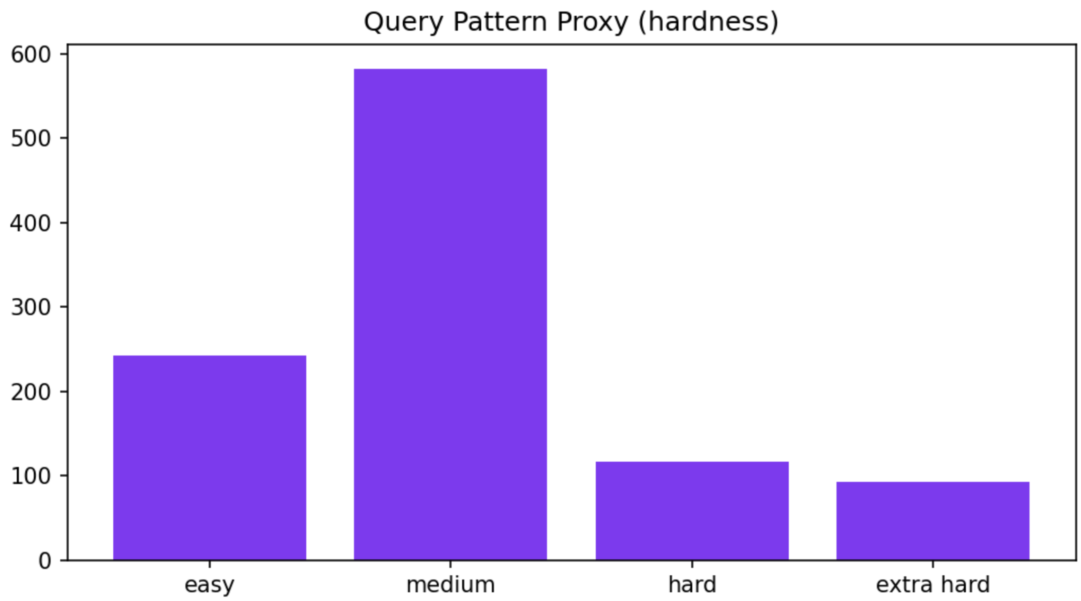

Figure 5: Structural hardness proxy (no NL available locally)

9. Database Schema Analysis

Across 166 schemas: avg 5.28 tables/DB (median 4.0, max 26), avg 27.13 columns/DB (median 19.5, max 352), avg 4.79 foreign keys/DB.

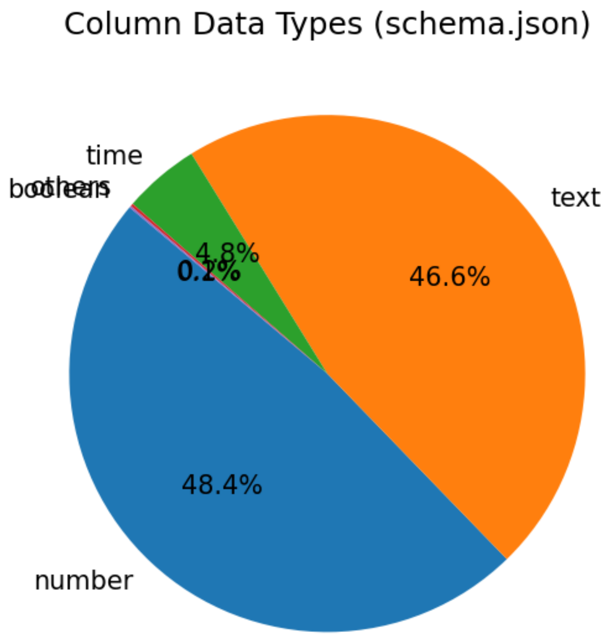

Figure 6: Column data type distribution

Type	Count	%
number	2178	48.4%
text	2097	46.6%
time	215	4.8%
others	8	0.2%
boolean	5	0.1%

9.1 Domain Distribution (Heuristic)

Domain	DB Count
other	95
business	22
education	13
entertainment	10
transport	10
sports	7
geography	3
food	3
science	2
healthcare	1

10. SQLite Database Profiling

Profiled 166 databases (166 with readable SQLite files).
Tables/DB: min 2, max 26, mean 5.3.
Rows/DB: min 0, max 553,693, mean 9,665, median 38.

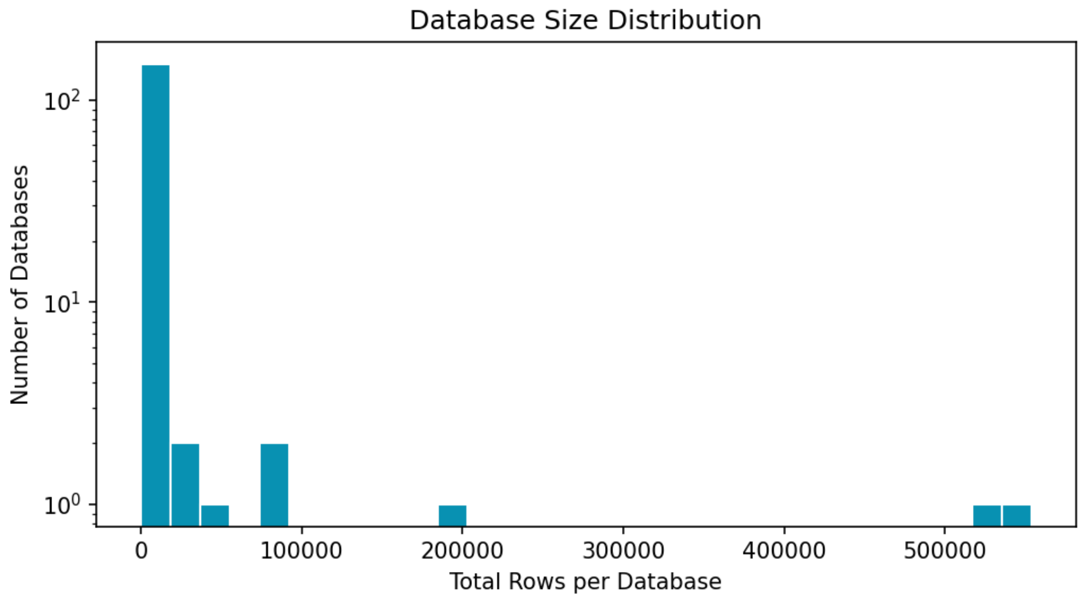

Figure 7: Database row counts

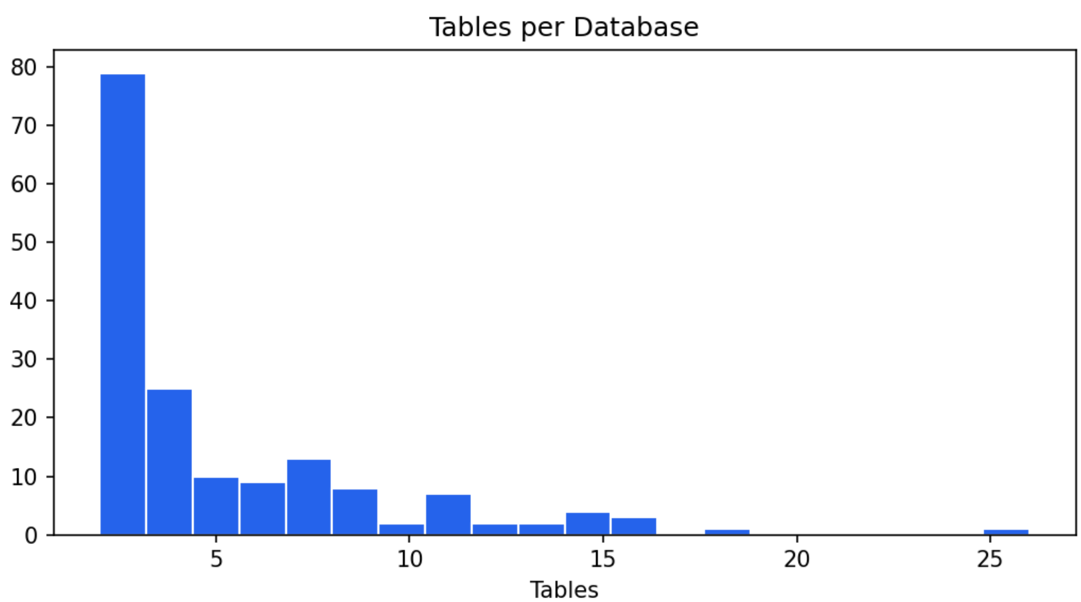

Figure 7b: Tables per database

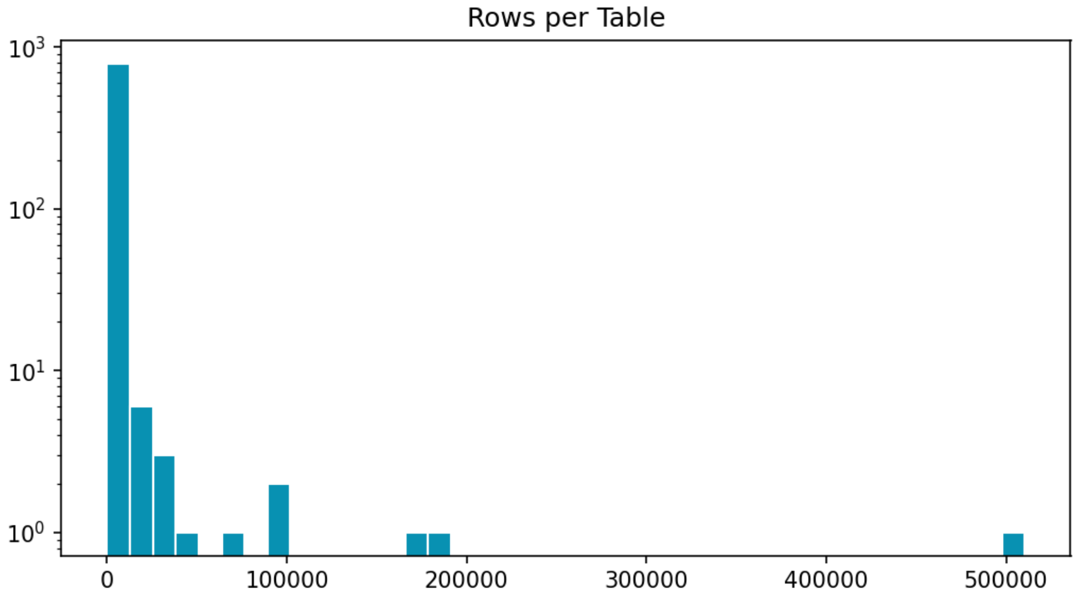

Figure 7c: Rows per table

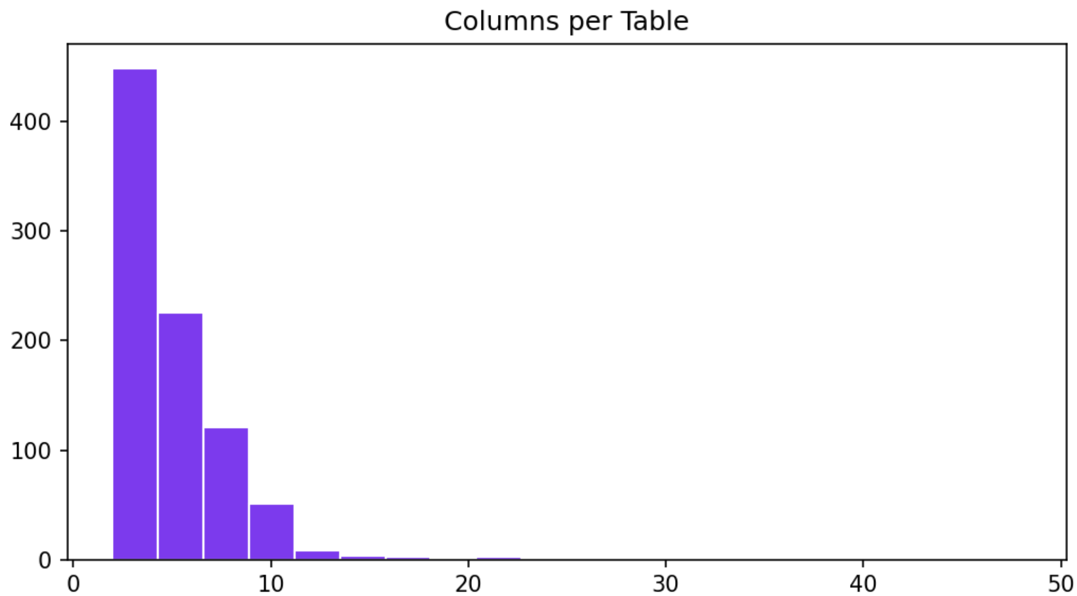

Figure 7d: Columns per table

Database	Tables	Rows	Columns	NULL %	FKs
baseball_1	26	553,693	352	0.0	19
wta_1	3	531,377	43	0.0	3
soccer_1	6	196,817	84	0.62	5
formula_1	13	88,380	94	0.0	17
flight_4	3	80,586	24	8.26	3
sakila_1	16	46,263	89	0.41	21

college_2	11	34,620	46	0.0	20
bike_1	4	22,181	46	0.0	1

11. Data Quality Assessment

- SQLite files found/opened: 166 / 166
- Mean NULL % per DB: 1.55% (max 45.55%)
- Columns with NULLs: 178 / 4497
- Empty databases (0 rows): 7
- Tables profiled: 873
- No remote downloads were used; analysis is fully offline from this directory.

12. Sample Gold SQL Examples

Database	Tier	Hardness	JOINs	SQL (truncated)
concert_singer	simple	medium	0	SELECT count(*) FROM singer
concert_singer	simple	medium	0	SELECT count(*) FROM singer
concert_singer	simple	easy	0	SELECT name , country , age FROM singer ORDER BY age DESC
concert_singer	simple	easy	0	SELECT name , country , age FROM singer ORDER BY age DESC
concert_singer	simple	medium	0	SELECT avg(age) , min(age) , max(age) FROM singer WHERE country = 'France'
concert_singer	simple	medium	0	SELECT avg(age) , min(age) , max(age) FROM singer WHERE country =

				'France'

13. Text-to-SQL Pipeline Design Recommendations

Schema input: Serialize tables/columns/types/FKs from schema.json (+ SQLite PRAGMA for runtime).

JOIN modelling: 39.5% of local gold queries use JOIN; focus on INNER JOIN + FK paths.

FK supervision: 93.8% of parsed JOIN ON pairs align with schema FKs.

Evaluation: Prefer execution accuracy on SQLite over string match.

Exports: Use eda_csvs/table_analysis.csv and join_analysis CSVs for debugging schema linking.

14. Preprocessing Completed

- Profiled 166 SQLite databases
- Exported database_summary.csv, table_analysis.csv, column_summary.csv
- Exported join_analysis.csv, join_conditions.csv, table_cooccurrence.csv, schema_fk_graph.csv, query_relations.csv
- Classified 1034 gold SQL queries
- Generated 24 charts under figures/

15. Data Governance

- Source: Local Spider snapshot in this directory
- License: Academic/research use (original Spider terms)
- Privacy: Synthetic / anonymized research DBs
- Reproducibility: python generate_milestone2_report.py (or eda_spider.py)

16. Deliverables

Deliverable	Description
Deep SQL EDA Report.docx	This EDA report (regenerated)
eda_results.json	Machine-readable statistics
eda_csvs/database_summary.csv	Per-database profiles
eda_csvs/table_analysis.csv	Per-table profiles
eda_csvs/column_summary.csv	Per-column NULL/cardinality
eda_csvs/query_relations.csv	Per-query SQL structure
eda_csvs/join_analysis.csv	Per-DB JOIN rates
eda_csvs/join_conditions.csv	Parsed JOIN ON pairs + FK flags
eda_csvs/table_cooccurrence.csv	Table pair co-occurrence

eda_csvs/schema_fk_graph.csv	Schema FK utilization
figures/	24 visualization charts
generate_milestone2_report.py	Reproducible EDA generator

17. Planned Activities for Milestone 3

- Add full train/dev/test JSON if available for NL question-pattern EDA
- Implement schema serialization for model prompts
- Build execution-based evaluator on local SQLite DBs
- Train/evaluate baseline Text-to-SQL model
- Error analysis by JOIN / aggregation / wrong-table categories

18. Summary

Offline EDA covered 166 schemas and 1034 gold SQL queries. JOIN prevalence is 39.5%, aggregations 53.3%, subqueries 15.4%. FK alignment on parsed JOIN conditions is 93.8%. Mean NULL rate across DBs is 1.55%. CSV exports under eda_csvs/ capture table and JOIN findings for Milestone 3.

Appendix: Deep SQL EDA Findings

Advanced structural analysis of local gold SQL: JOIN types, hardness, subqueries, WHERE operators, table references, FK alignment, and relation topology.

1. Overview

Analysed 1034 SQL queries. JOIN count distribution: {0: 626, 1: 320, 2: 72, 3: 10, 4: 6}.

2. JOIN Analysis

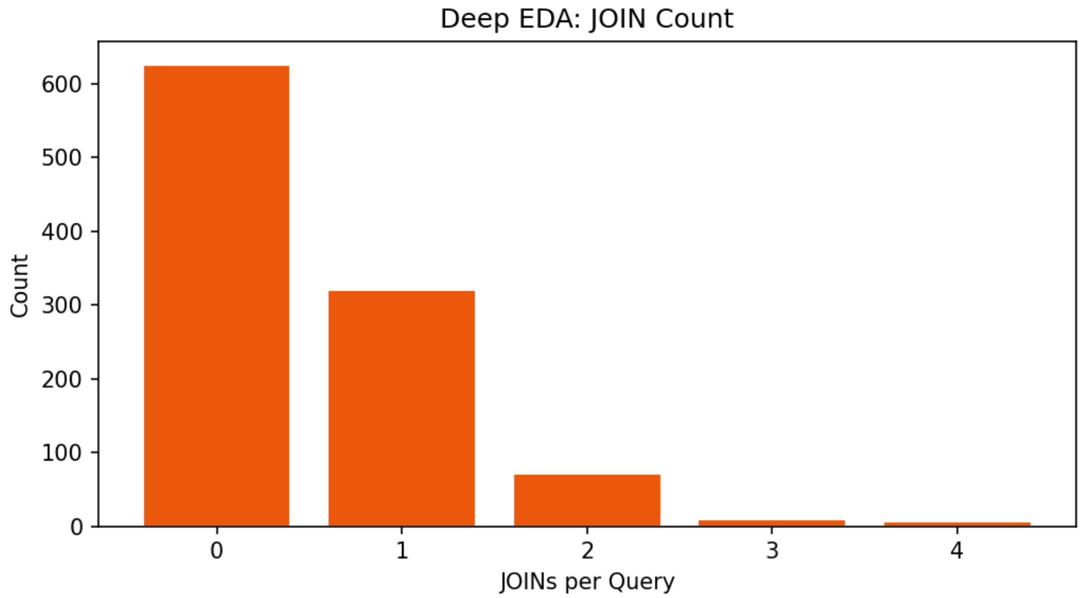

Figure A1: JOIN count distribution

39.5% of queries contain at least one JOIN. 88 queries have 2+ JOINs.

2.1 JOIN Types

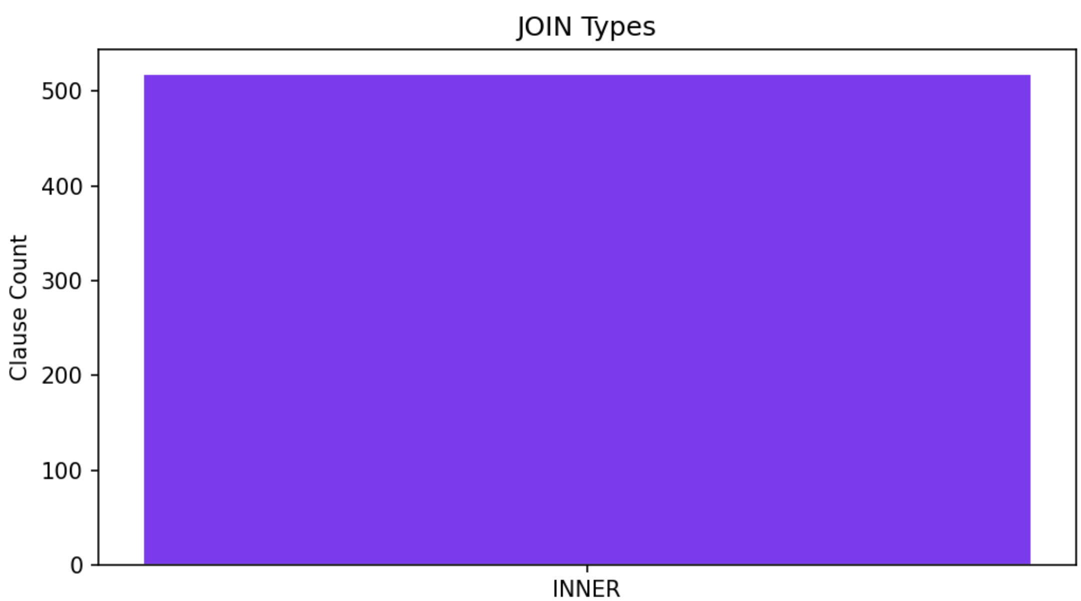

Figure A2: JOIN types

JOIN type	Count
INNER	518

Spider typically uses bare JOIN (INNER). LEFT/RIGHT/CROSS are rare or absent.

2.2 Databases with Highest JOIN Rates

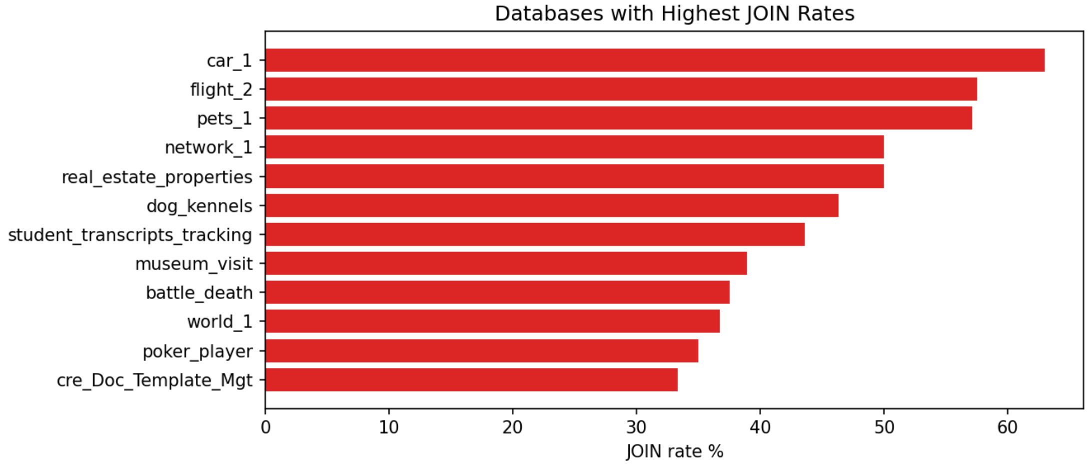

Figure A3: JOIN rate by database

Database	Queries	With JOIN	JOIN %
car_1	92	58	63.0
flight_2	80	46	57.5
pets_1	42	24	57.1
network_1	56	28	50.0
real_estate_properties	4	2	50.0
dog_kennels	82	38	46.3
student_transcripts_tracking	78	34	43.6
museum_visit	18	7	38.9
battle_death	16	6	37.5
world_1	120	44	36.7
poker_player	40	14	35.0
cre_Doc_Template_Mgt	84	28	33.3

3. Subquery Analysis

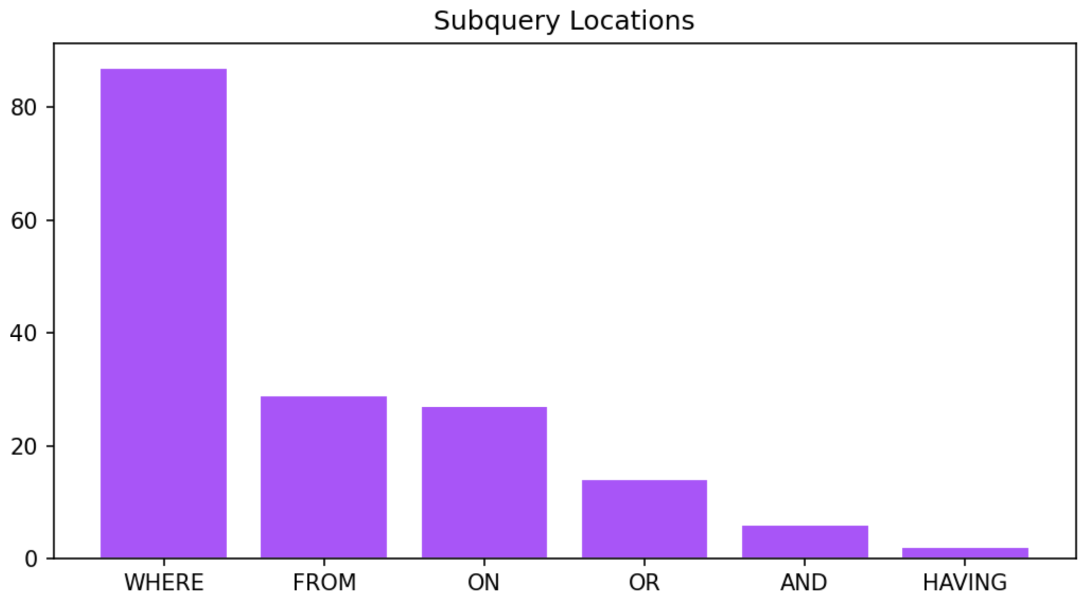

Figure A4: Subquery locations

Subquery prevalence: 15.4%. Locations: {'WHERE': 87, 'FROM': 29, 'ON': 27, 'OR': 14, 'AND': 6, 'HAVING': 2}.

4. Spider Official SQL Hardness

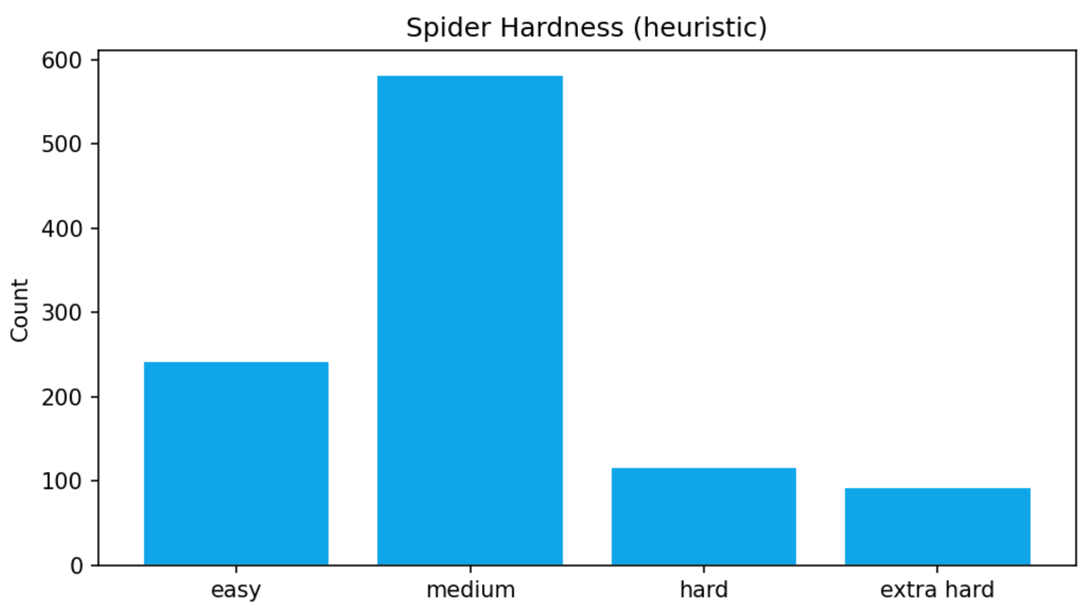

Figure A5: Hardness distribution

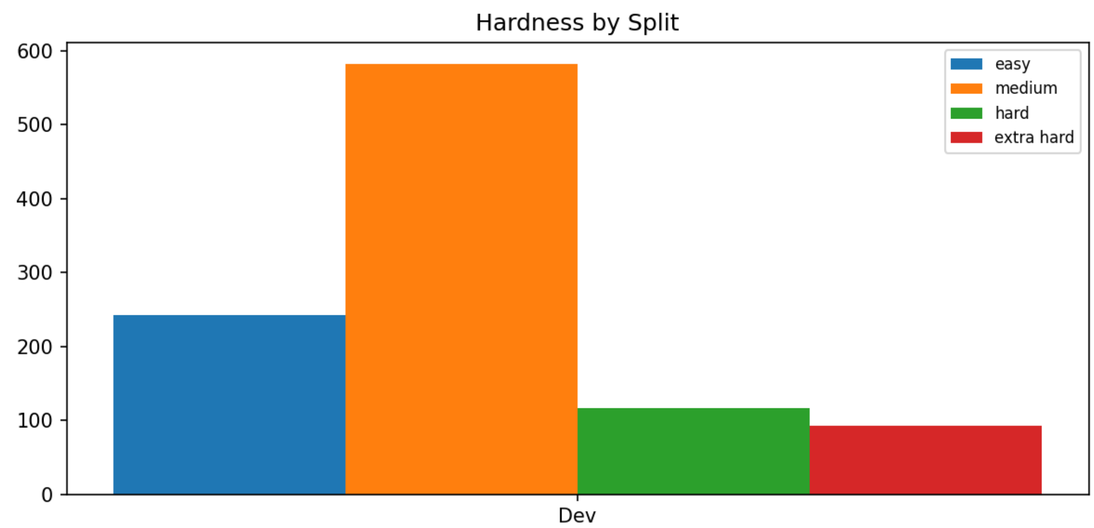

Figure A6: Hardness by split

Hardness	Count
medium	582
easy	242
hard	117
extra hard	93

5. Table References per Query

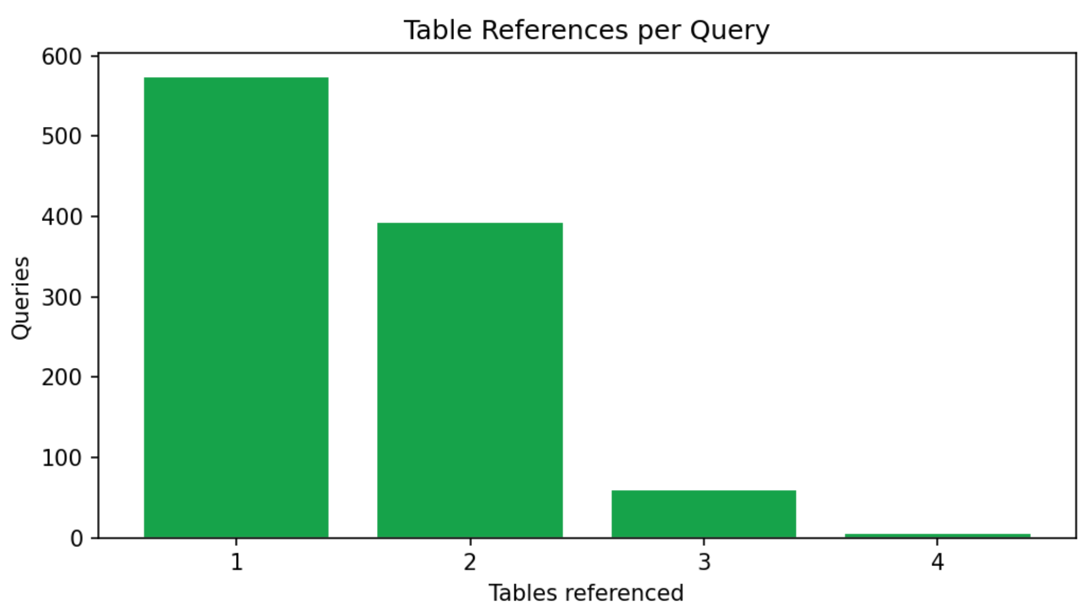

Figure A7: Tables referenced

# Tables	Queries

1	575
2	393
3	60
4	6

6. WHERE Clause Analysis

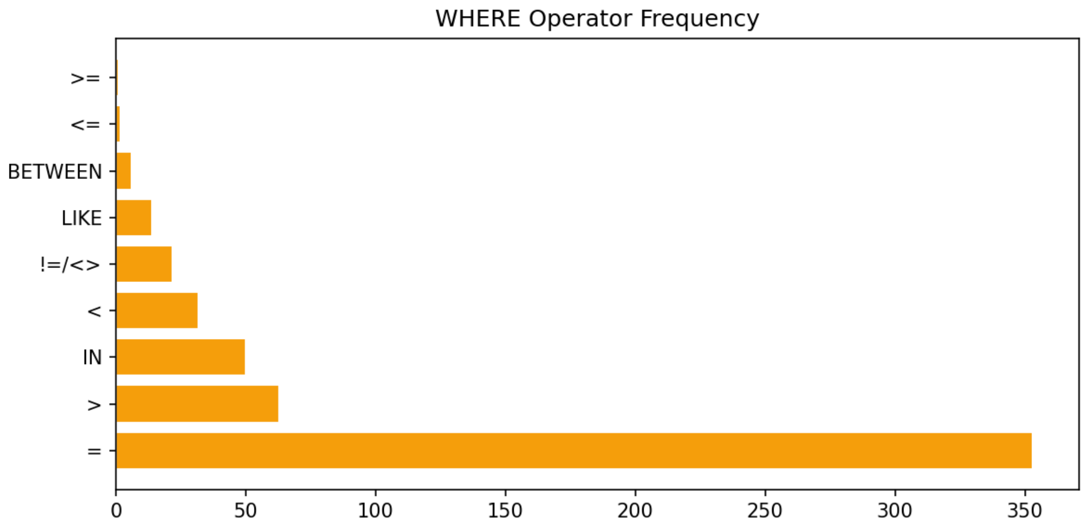

Figure A8: WHERE operators

Operator	Queries containing it
=	353
>	63
IN	50
<	32
!=/<>	22
LIKE	14
BETWEEN	6
<=	2
>=	1

7. Aggregation Analysis

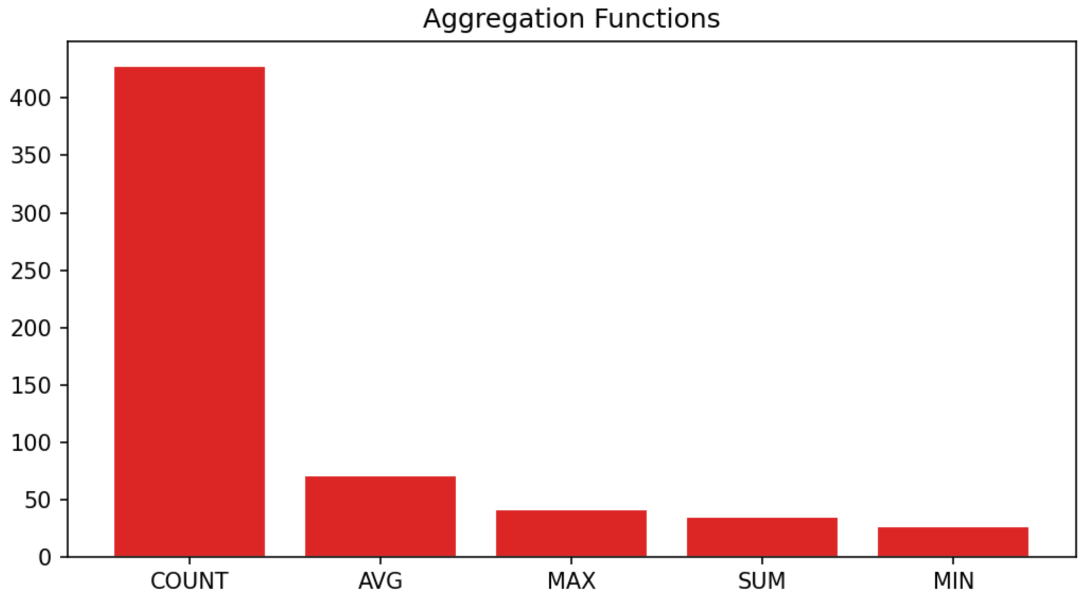

Figure A9: Aggregations

Function	Count
COUNT	428
AVG	71
MAX	42
SUM	35
MIN	27

8. SELECT / Other Clauses

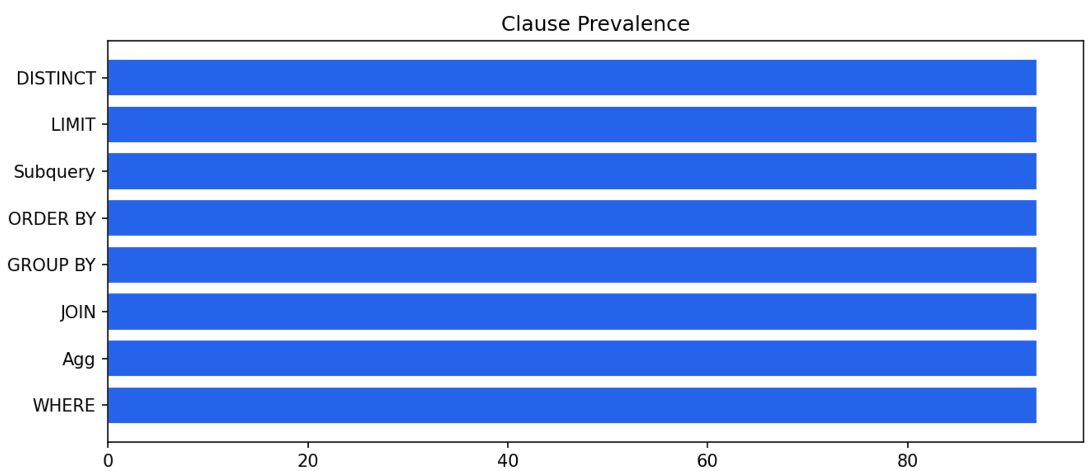

Figure A10: Clause prevalence

9. Query Relations Analysis

9.1 Relation Topology

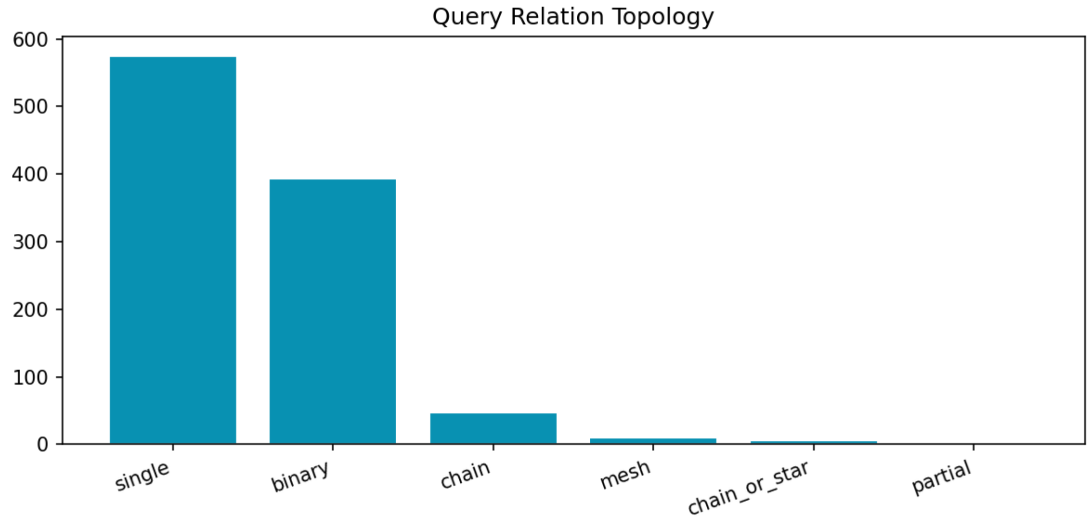

Figure A11: Topology

Topology	Count
single	575
binary	393
chain	47
mesh	10
chain_or_star	6
partial	3

9.2 Foreign Key Alignment in JOINs

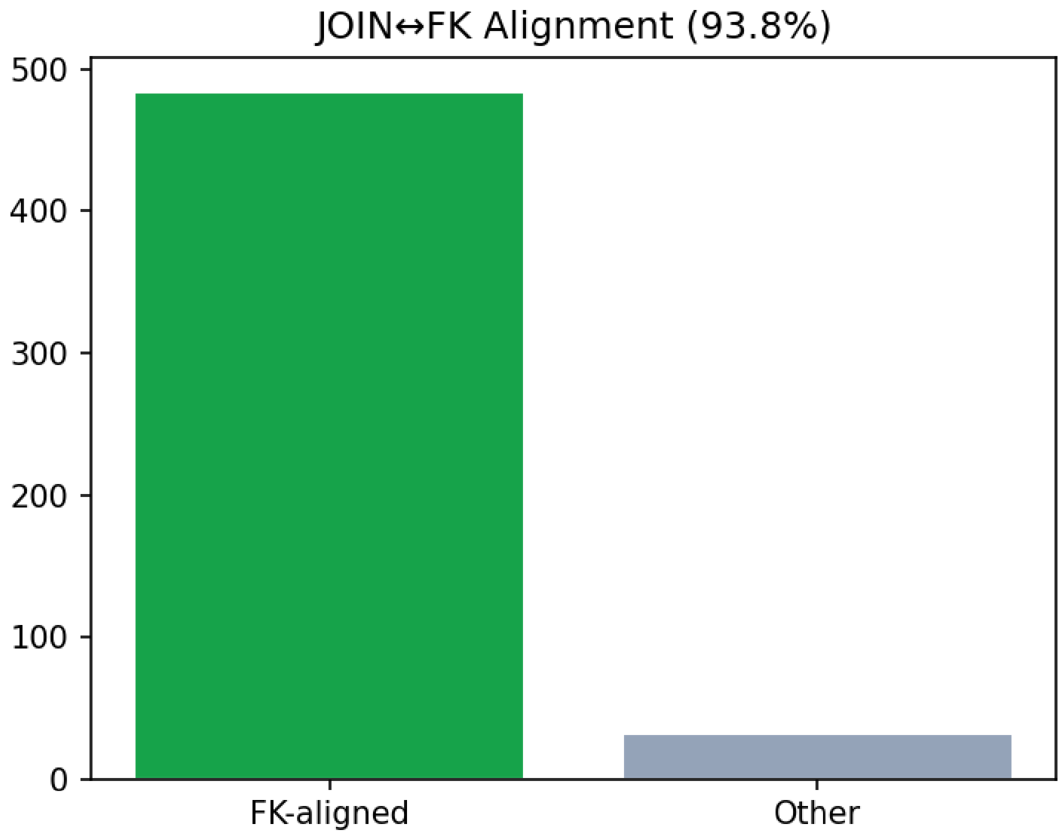

Figure A12: FK alignment

Of 516 parsed JOIN ON column pairs, 484 (93.8%) match schema foreign keys.

9.3 Table Co-occurrence & JOIN Patterns

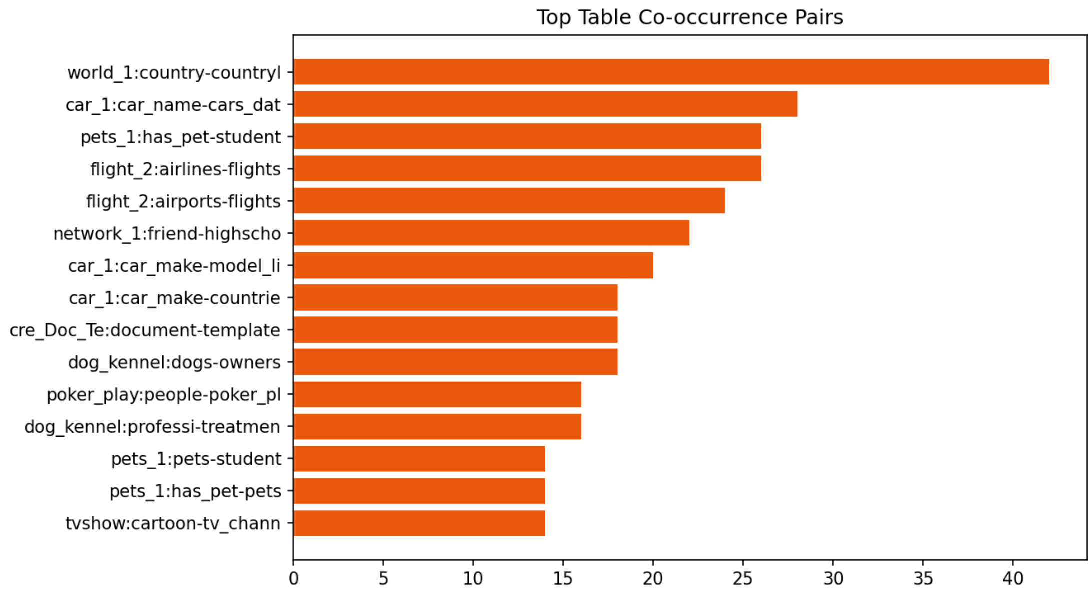

Figure A13: Top table pairs

Database	Table A	Table B	Co-occurrence
world_1	country	countrylanguage	42
car_1	car_names	cars_data	28
pets_1	has_pet	student	26
flight_2	airlines	flights	26
flight_2	airports	flights	24
network_1	friend	highschooler	22
car_1	car_makers	model_list	20
car_1	car_makers	countries	18
cre_Doc_Template_Mgt	documents	templates	18
dog_kennels	dogs	owners	18
poker_player	people	poker_player	16
dog_kennels	professionals	treatments	16
pets_1	pets	student	14
pets_1	has_pet	pets	14
tvshow	cartoon	tv_channel	14

9.4 Schema FK Utilization by Database

Database	Tables	FKs	FK coverage %
hospital_1	15	25	100.0
cre_Drama_Workshop_Groups	18	25	100.0
sakila_1	16	21	93.8
college_2	11	20	90.9
baseball_1	26	19	73.1
academic	15	18	93.3
assets_maintenance	14	18	100.0

formula_1	13	17	84.6
local_govt_and_lot	11	15	100.0
customer_deliveries	13	15	100.0
cre_Theme_park	16	14	100.0
department_store	14	13	100.0

10. Key Takeaways for Text-to-SQL

- 39.5% of local gold queries need JOINs — schema linking is mandatory.
- JOIN types observed: {'INNER': 518} — model INNER JOIN first.
- 93.8% of JOIN ON pairs align with schema FKs — use FKs as join supervision.
- Subqueries in 15.4% of queries — mostly WHERE/SELECT nested SELECTs.
- Dominant aggregation: COUNT.
- Evaluate with execution accuracy so equivalent JOIN order still passes.
- See eda_csvs/ for full per-table and per-JOIN exports.

Signatures

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